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By developing a high-performance computing enabled **probabilistic model** for crop flowering time, we show that existing genotypic variation in thermal response of sunflower flowering time predicts **agronomically divergent patterns** in future environments.

Helianthus annuus (domestic sunflower) is an important crop for biofuels, oils, livestock feed, and a promising cash crop for farmers in warmer, drier regions. Recent research from our team revealed that flowering time (**phenology**) alone accounts for roughly 9% of all variation in yield and pinpointed several promising candidate genes related to flower timing in commercially grown sunflower varieties (Clark et al. 2025; Elmendorf et al. in review; Innes et al. in prep)

Research Questions

- (Q1) Is genotypic variation in phenology attributable to differences in **optimum developmental temperature** / developmental **thresholds**?
- (Q2) Does this variation (or lack thereof) imply **agronomically meaningful** crop flowering time differences in future environments?

By answering these questions, we move from purely phenomological patterns into actionable breeding targets for commercial or governmental crop breeders as well as process insight for farmers.

Methods

We developed a generative model (Fig. 1) for crop flowering time to estimate genotypic variation in latent phenological thermal response curves (Fig. 2). To this end, we modeled sunflower flowering time as a latent continuous developmental process in a time-to-event framework. Flowering onset for each individual was treated as occurring only once this unobservable development state reached a genotype specific threshold (Fig. 3) following Betancourt et al (2024).

Instantaneous development rate as a function of temperature was represented using a Wang-Engel nonlinear response function (Wang & Engel, 1998). The Wang-Engel function rises from a minimum threshold temperature (T_{min}) to an optimum (T_{opt}) and declines toward a maximum (T_{max}). This asymmetric, unimodal form captures the biologically realistic acceleration and deceleration of development around an optimal temperature.

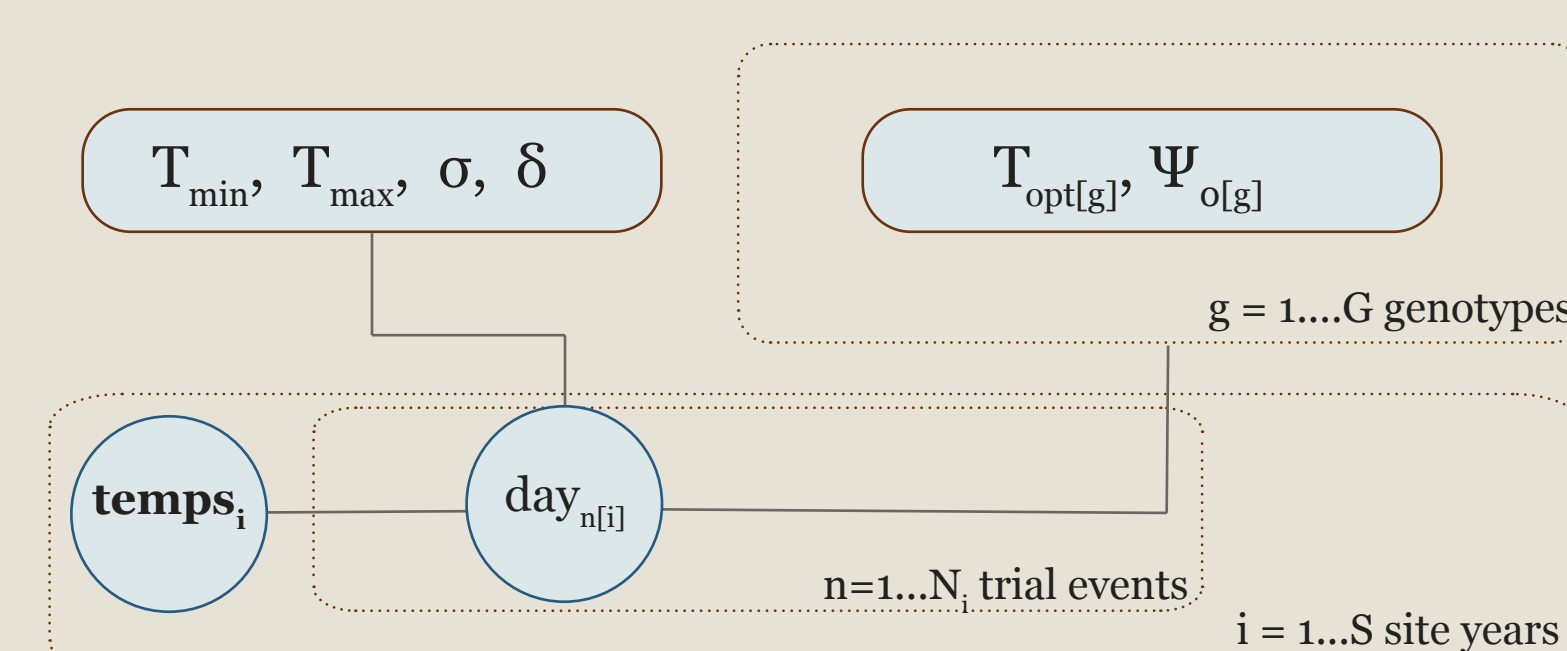


Figure 1. Bayesian network diagram for the probabilistic phenology model. Plate diagram for the developed model demonstrates the conditional independencies across site years given global and genotypic parameters, enabling within-chain parallelization via **multithreading**.

Figure 2. Genotypes have unique temperature response curves with varying optimal development temperatures. Some varieties may develop faster in warmer or cooler environments.

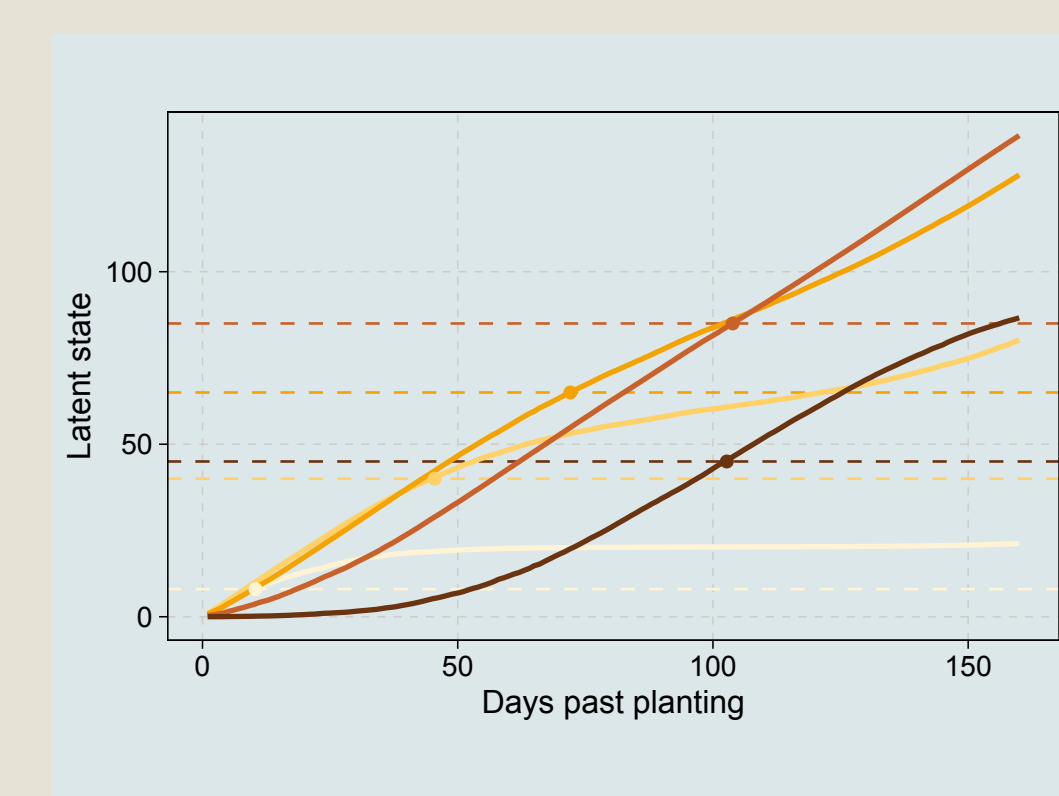


Figure 3. Genotypes also vary in their development thresholds. Differences among genotypes in development thresholds (horizontal dashed lines) interacts with latent state trajectories.

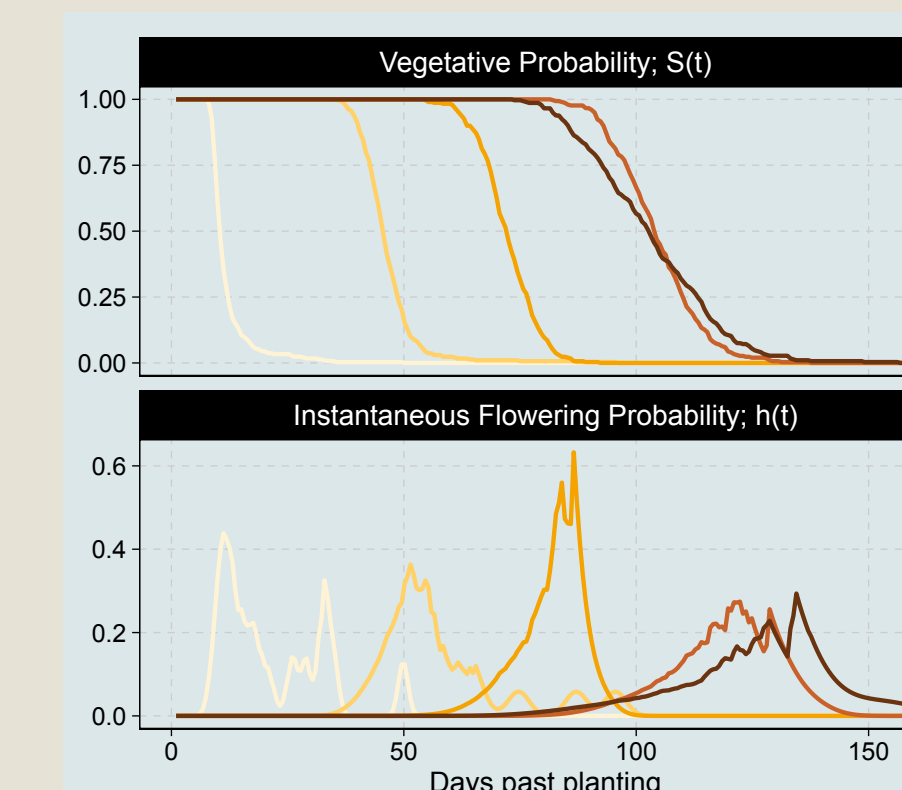


Figure 4. Emergent genotypic variation in time to flower. As a result of genotypic variation in temperature sensitivity (Fig. 2) and development thresholds (Fig 3), the model captures emergent variation in time to flower (vertical dotted lines).

Genotype-specific thermal response parameters were estimated within a hierarchical Bayesian framework via the No-U-Turn Sampler for Hamiltonian Monte Carlo implemented in Stan (Fig. 1). The model was fit jointly to repeated-observation ($n=11,610$) phenological records collected across more than **670 unique genotypes** and dozens of field environments, with daily temperature forcing driving the accumulation of developmental progress toward flowering.

Fitted genotype-specific thermal response curves were subsequently projected onto observed and future temperature regimes using moderate projections from the latest Community Earth System Model (CESM) data to evaluate whether genotypes with similar historical flowering times diverge in projected flowering behavior under warming.

Role of high-performance computing

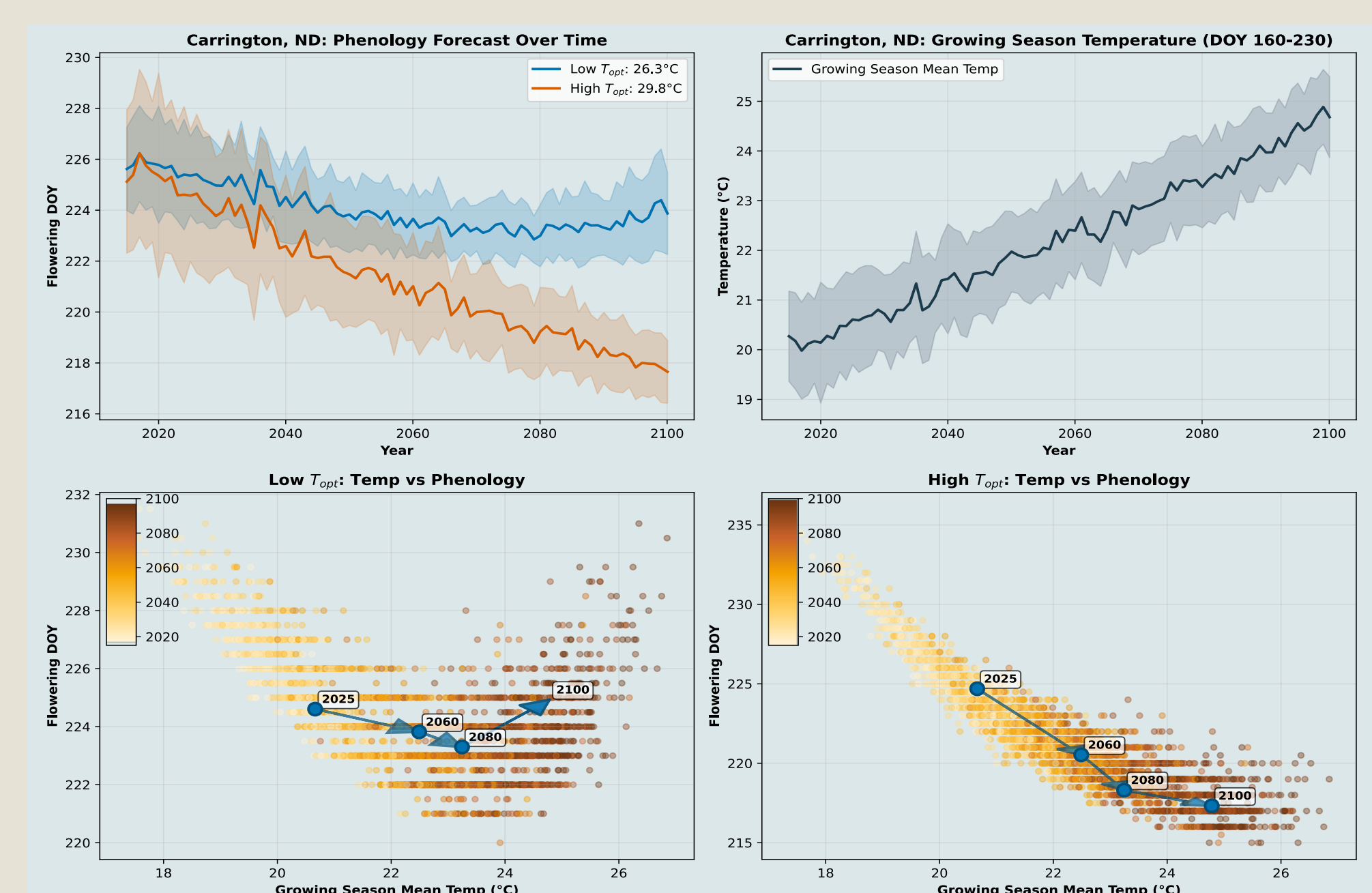
Scalable inference required HPC at two stages:

- Fitting this model required sampling from a very high-dimensional joint posterior; this model is intractable on a laptop. We exploited the conditional independencies in the likelihood to enable **within-Markov-chain parallelization via multithreading** on a 72-thread CPU, making posterior inference tractable.
- Parallelized ensemble inference used Numba's **JIT compilation**, enabling batch forecasting across genotypes and CESM ensemble members with worker functions designed for multiprocessing dispatch via SLURM.

Results

- Across all 670+ genotypes, optimum developmental temperature varied by 3°C (Q1).
- Phenological forecasts for the genotypes with the lowest and highest T_{opt} values at a representative farm in Carrington, ND show strong divergences by the end of the century (Figure 5) (Q2).
- At Carrington, mean growing season temperatures are expected to increase by $3.5\text{--}5.5^{\circ}\text{C}$ by 2100. As a result, lower optimum temperature genotypes (blue, top left) will advance their flowering date until about 2060, at which point flowering time begins to be delayed as growing season temperatures become supraoptimal. In contrast, higher optimum temperature genotypes (orange, top left) will continue to advance, flowering up to 10 days earlier than historical mean flowering times.

Figure 5. Genotypic variation in optimum development temperature predicts divergent phenological trajectories under future environments. Phenology forecasts to 2100 for both low (blue) and high (orange) genotypes (top left) in a representative farm under future temperature regimes (top right). Flowering day of year given temperature for 1000 posterior draws & 50 CESM LENS2 ensemble members for both genotypes (bottom row).



Discussion

- Genotypic variation in optimum developmental temperature (T_{opt}) across the SAM panel was modest ($\sim 3^{\circ}\text{C}$) but sufficient to drive divergent, not merely offset, phenological trajectories under warming, directly answering our two motivating questions.
- This divergence emerges because development rate does not increase monotonically with temperature: once growing-season conditions exceed a genotype's T_{opt} , further warming slows rather than accelerates flowering.
- Notably, the lowest- T_{opt} genotype was bred in North Dakota and the highest- T_{opt} genotype in Texas, suggesting breeding programs may already be selecting on thermal response shape rather than mean flowering time alone. These results indicate that T_{opt} is an actionable breeding target distinct from flowering time itself, and that genotype-specific thermal response curves could improve variety-selection tools as growing environments change.

References & Acknowledgements

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